Bayesian Analysis of Clustered Semi-Competing Risks

Application to Multi-Center Breast Cancer Data

Jinheum Kim

Department of Applied Statistics, University of Suwon

33rd International Biometric Conference (IBC 2026)

July 12–16, 2026 Seoul, South Korea

Outline

Why clustered semi-competing risks?

Multi-center cancer cohorts are three things *at once*:

- **Multi-state**: each patient moves through diagnosis $\rightarrow$ relapse $\rightarrow$ death.
- **Semi-competing**: relapse can be censored by death, but not the reverse.
- **Clustered**: patients are grouped by contributing center.

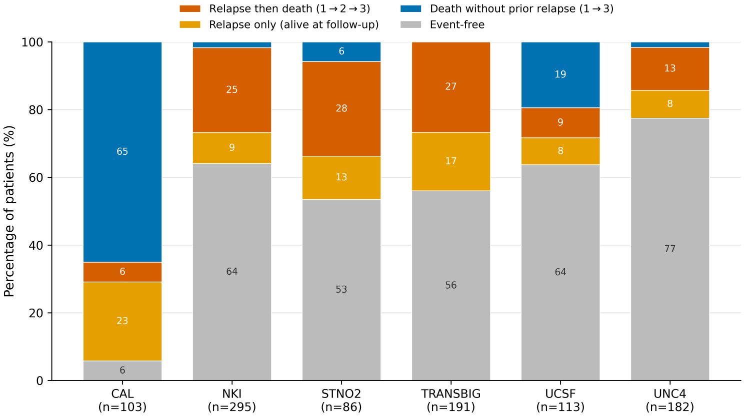
Centers differ beyond measured covariates, and ignoring this underestimates uncertainty. The standard remedy: a **center-level frailty**, usually shared across transitions. But a center's excess risk in relapse need not match its risk in death.

The gap

Existing work sits at two extremes: shared frailty, or fully correlated multivariate-normal frailty. The in-between case, transition-specific and independent, is rarely compared with these two, and there's no clear rule for choosing among them.

This talk. We treat all three structures as competing candidates, fit by Hamiltonian Monte Carlo (HMC) and compared by leave-one-out cross-validation (LOO-CV), letting a simpler structure win when the data don't need more.

Outcome proportions differ across centers



Six breast-cancer cohorts: outcome-category shares differ across centers (e.g. CAL: 65% death without relapse; UNC4: 77% event-free) and within centers.

Observed data and notation

Clustered multi-state data (long format). For each subject $i = 1, \ldots, N$,

$$\mathcal{D} = \left\{ D_{ij} = (T_{ij}, \Delta_{ij}, R_{ij}, G_i, X_i) \right\}_{j=1}^{n_i},$$

- T_{ij}: stop time of the j-th transition interval; $\Delta_{ij} \in \{0, 1\}$: event indicator.
- $R_{ij} \in \{1, \ldots, T\}$: transition type; $G_i \in \{1, \ldots, K\}$: cluster (center).
- $X_i \in \mathbb{R}^P$: covariates, shared across transitions; n_i: number of records for subject i.

Semi-competing risks (illness–death) is the special case with $T = 3$:

$$1 \to 2 \text{ (relapse)}, \quad 1 \to 3 \text{ (death w/o relapse)}, \quad 2 \to 3 \text{ (death after relapse)}.$$

The model: transition-specific hazards with a center frailty

Transition-specific Cox hazard with a cluster frailty U, for the three transitions defined above ($t = 1, \dots, T$):

$$h_t(u \mid X_i, G_i) \;=\; h_{0t}(u)\; U_{G_i,t}\, \exp(X_i^\top \beta_t),$$

where $U_{g,t} > 0$ acts multiplicatively on the hazard for cluster $g = G_i$ at transition t: $U_{g,t} > 1$ signals *elevated* risk, $U_{g,t} < 1$ *reduced* risk.

- Because subjects in cluster g share the same unobserved $U_{g,t}$, a standard Cox fit misses the resulting **within-cluster dependence**. Adding the frailty term restores it.
- $\mathrm{Var}(U_{g,t})$ (denoted θ_t or τ_t, next slide) measures **between**-cluster heterogeneity, *not* within-cluster variation. Shared vs. transition-specific is the question ahead.

Three frailty specifications

Model	Frailty structure	Captures
Shared-γ	$U_g \sim \Gamma(1/\theta, \theta)$, one per cluster, common to all transitions	overall cluster heterogeneity (1 variance θ)
TS-γ	$U_{gt} \sim \Gamma(1/\theta_t, \theta_t)$, separate frailty per transition, independent across t	transition-specific heterogeneity $(\theta_1, \theta_2, \theta_3)$
TS-MVN	$(\log U_{g1}, \log U_{g2}, \log U_{g3}) \sim \mathcal{N}_3(0, \Sigma)$	transition-specific heterogeneity *and* between-transition correlation

All share $E(U) = 1$ (mean-one frailty), so β_t stays interpretable as a log-hazard ratio.

Gaussian-variance variants. **Shared-normal** and **TS-normal** replace the gamma frailty with log-normal; lacking between-transition correlation, they behave like their gamma counterparts and are folded into the comparison rather than shown separately.

Frequentist estimation: penalized partial likelihood

Step 1: Conditional partial likelihood (given the frailties U):

$$L(\beta, U) = \prod_{i,j:\, \Delta_{ij}=1} \frac{U_{G_i,R_{ij}} \exp(X_i^\top \beta_{R_{ij}})}{\sum_{\ell \in \mathcal{R}_{R_{ij}}(T_{ij})} U_{G_\ell,R_{ij}} \exp(X_\ell^\top \beta_{R_{ij}})}.$$

where the risk set for transition t at time u is $\mathcal{R}_t(u) = \{\ell : \; R_{\ell j} = t, \; T_{\ell,j-1} < u \le T_{\ell j}\}$.

Step 2: Integrate out the frailty against its distribution $f(U \mid \theta)$:

$$L_{\text{marg}}(\beta, \theta) = \int L(\beta, U) \prod_g f(U_g \mid \theta)\, dU.$$

Step 3: Penalized partial log-likelihood

$$\ell_{\text{PPL}}(\beta, \theta) = \log L_{\text{marg}}(\beta, \theta), \qquad (\hat\beta, \hat\theta) = \arg\max_{\beta,\theta} \; \ell_{\text{PPL}}(\beta, \theta).$$

- Integrating out U_g shrinks cluster effects toward 1 and controls heterogeneity through the variance θ (Ripatti & Palmgren, 2000).
- Ties handled by Efron's method; inference via asymptotic SE from the curvature of ℓ_{PPL}.

Estimation: frequentist (PPL) vs. Bayesian (HMC)

The *same* frailty model is fit two ways; we contrast both.

	Frequentist (PPL)	**Bayesian (Stan/HMC)**
Target	point estimates $\hat{\beta}_t, \hat{\theta}$	posterior $p(\beta_t, U, \theta \mid \mathcal{D})$
Frailty	integrated out / penalized	sampled jointly with β_t, θ
Engine	maximize $\ell_{\text{PPL}}(\beta, \theta)$	NUTS draws from the joint posterior
Uncertainty	asymptotic SE (curvature / sandwich)	posterior SD, credible intervals

Bayesian posterior. With priors $\beta_{t,p} \sim N(0, 1)$ and $\theta \sim \Gamma(a_\theta, b_\theta)$,

$$p(\beta, U, \theta \mid \mathcal{D}) \;\propto\; \underbrace{L(\beta, U \mid \mathcal{D})}_{\text{same Cox partial likelihood}} \cdot \Big[\textstyle\prod_{t,p} \phi(\beta_{t,p}) \Big] \cdot \Big[\textstyle\prod_g f(U_g \mid \theta) \Big] \cdot p(\theta).$$

Why HMC? The risk-set denominator couples every subject's frailty, so none of β, U_g, θ has a closed-form conditional (non-conjugacy); HMC/NUTS explores the correlated posterior directly.

Simulation design

Scenarios (data-generating frailty structure):

- **None**: no frailty **Shared**: shared frailty
- **Indep**: transition-specific, independent **Corr**: transition-specific, correlated

Models compared (of five fit; three shown here):

- **Shared-**γ: one gamma frailty, shared across transitions.
- **TS-**γ: transition-specific gamma frailties.
- **TS-MVN**: transition-specific, correlated (multivariate-normal) frailties.

Metrics: bias, 95% coverage probability (CP), and SD ratio (SDR $=$ mean posterior SD / Monte Carlo SD; 1 means correct calibration).

Data-generating scenarios

Common design across all four scenarios:

- Same illness-death (3-transition) structure, same covariates (Z_1, Z_2), same true regression coefficients β.
- $N = 500$ subjects, $K = 20$ clusters per replicate; $R = 300$ Monte Carlo replications.
- Posterior inference via Stan/HMC (non-conjugate, no closed form): 1 chain, 500 warm-up + 500 sampling iterations (500 posterior draws).

Only the frailty variance structure changes across scenarios:

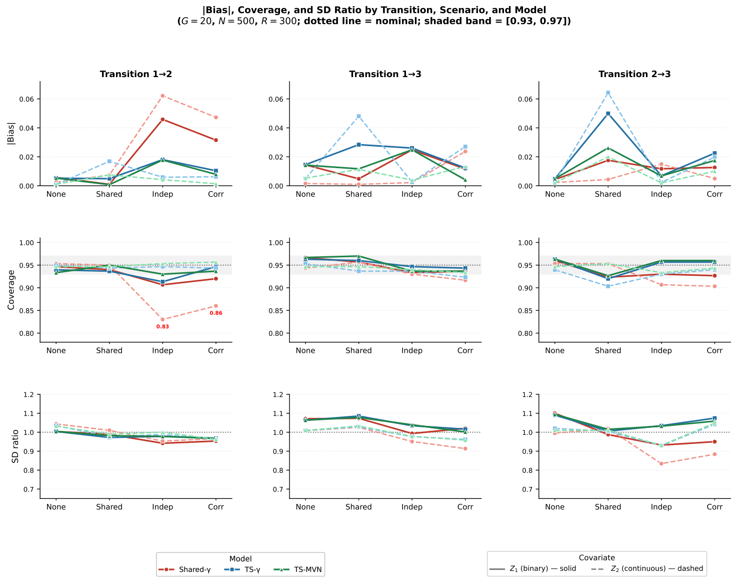
Scn	Frailty structure	True variance	True correlation (ρ)
None	none	0	—
Shared	shared across transitions	$\theta = 0.50$	—
Indep	transition-specific, independent	$\tau = 0.64$ (all trans.)	0
Corr	transition-specific, correlated	$\tau = 0.64$ (all trans.)	0.50, 0.30, 0.10

Results: coverage and calibration (numbers)

		$1 \to 2$				$1 \to 3$				$2 \to 3$			
		Z_1		Z_2		Z_1		Z_2		Z_1		Z_2	
Scn	Model	CP	SDR	CP	SDR	CP	SDR	CP	SDR	CP	SDR	CP	SDR
None	Shared-γ	0.95	1.00	0.95	1.04	0.97	1.07	0.94	1.01	0.96	1.10	0.95	0.99
	TS-γ	0.94	1.00	0.95	1.03	0.96	1.06	0.95	1.01	0.96	1.09	0.94	1.02
	TS-MVN	0.93	1.00	0.95	1.03	0.97	1.06	0.95	1.01	0.96	1.09	0.95	1.01
Shared	Shared-γ	0.94	0.99	0.95	1.01	0.96	1.07	0.96	1.03	0.92	0.99	0.95	1.02
	TS-γ	0.94	0.97	0.94	0.98	0.96	1.09	0.94	1.03	0.92	1.01	0.90	1.00
	TS-MVN	0.95	0.98	0.95	0.99	0.97	1.08	0.95	1.03	0.93	1.01	0.95	1.01
Indep	Shared-γ	0.91	0.94	**0.83**	0.95	0.93	0.99	0.93	0.95	0.93	0.93	0.91	0.83
	TS-γ	0.91	0.98	0.95	0.99	0.95	1.03	0.94	0.98	0.96	1.03	0.93	0.93
	TS-MVN	0.93	0.98	0.95	1.00	0.94	1.04	0.94	0.98	0.96	1.03	0.93	0.93
Corr	Shared-γ	0.92	0.95	**0.86**	0.97	0.94	1.02	0.92	0.91	0.93	0.95	0.90	0.88
	TS-γ	0.95	0.97	0.94	0.96	0.94	1.02	0.92	0.96	0.96	1.07	0.94	1.04
	TS-MVN	0.94	0.97	0.96	0.96	0.94	1.00	0.93	0.96	0.96	1.06	0.94	1.05

Z_1: binary (β=0.5), Z_2: continuous (β=1.0); SDR = mean posterior SD / MC SD. Red marks CP < 0.90.
$K = 20$ clusters, $N = 500$, $R = 300$; $R = 298$ for None$\times$TS-γ (2 extreme-runtime reps excluded).
Shared-γ under-covers at the $1 \to 2$ transition for Z_2 once frailty is transition-specific (Indep, Corr).

Results: the same story, visualized



Reading the grid

Rows: |Bias|, coverage, SD ratio. Columns: transitions $1 \to 2$, $1 \to 3$, $2 \to 3$.
Color = model; full tone = Z_1 (binary), light tone = Z_2 (continuous).

- Point estimation is broadly comparable; bias stays small across models and conditions.
- Shared-γ under-covers for Z_2 (continuous) at $1 \to 2$: CP$= 0.83$ (Indep), CP$= 0.86$ (Corr); SDR drops to 0.83–0.95.
- Coverage drop tracks SDR drop: narrower posterior SD $\Rightarrow$ lower coverage (a **calibration** failure, not a bias failure).
- TS-γ and TS-MVN stay near-nominal (≥ 0.91) across all transitions, scenarios, and covariates.

Results: frailty-variance recovery

Scale parameter (θ for gamma, τ for MVN)
reported only where the estimand matches the data-generating frailty family

Scn	Model	Param	True	Est	Bias	CP
None	Shared-γ	θ	0	0.010	—	**0**
	TS-γ	$\bar{\theta}$	0	0.033	—	**0**
	TS-MVN	$\bar{\tau}$	0	0.146	—	**0**
Shared	Shared-γ	θ	0.50	0.551	+0.051	0.96
	TS-γ	$\bar{\theta}$	0.50	0.524	+0.031	0.95
	TS-MVN	$\bar{\tau}$	0.50	0.530	+0.040	0.95
Indep	TS-MVN	$\bar{\tau}$	0.64	0.675	+0.038	0.94
Corr	TS-MVN	$\bar{\tau}$	0.64	0.666	+0.030	0.95

Between-transition correlation (TS-MVN, correlation scale ρ): Indep vs. Corr

Scn	ρ_{ts}	True	Est	Bias	CP
Indep	all three	0	≈ 0	≈ 0	$\geq$ nom.
Corr	ρ_{21}	0.50	0.178	-0.322	0.97
	ρ_{31}	0.30	0.083	-0.217	0.99
	ρ_{32}	0.10	0.017	-0.083	1.00

$\bar{\theta}, \bar{\tau}$: transition-averaged variances (equal by design); ρ_{ts}: posterior-mean correlations. Wide posteriors keep CP $\geq$ nominal despite shrinkage.

Reading the variance results

- **No-frailty boundary pathology** (true $= 0$): the frailty variance lies on the positive boundary, so an equal-tailed credible interval cannot contain 0. Coverage is therefore 0 *by construction*: a structural artifact of a positively constrained parameter, not an estimation failure.
- **Scale is otherwise well behaved**: where the estimand matches the data-generating family (Shared θ; Indep/Corr transition-specific τ), the variance is recovered with only a modest small-sample upward bias and near-nominal coverage.
- **Correlation is the hard part**: when the truth is independence (Indep, $\rho = 0$) the correlations are correctly estimated near 0; but *real* correlations (Corr) are strongly *shrunk toward* 0 (ρ_{21}: $0.50 \rightarrow 0.18$; ρ_{31}: $0.30 \rightarrow 0.08$; ρ_{32}: $0.10 \rightarrow 0.02$). The likelihood identifies the off-diagonals only weakly at this cluster count, so the LKJ prior pulls them toward 0. Wide posteriors keep coverage at or above nominal (over-coverage): a calibration–identifiability trade-off that mirrors, in reverse, the under-coverage seen for β.

Real-world application: breast cancer multi-center cohort

Data source. Six international breast cancer cohorts: CAL, NKI, STNO2, TRANSBIG, UCSF, UNC4 ($K = 6$ **clusters**).

Cohort construction.

- Source resource (Haibe-Kains et al.): 5,715 patients across 36 datasets.
- Six cohorts recording RFS, OS, and treatment status: 1,238 **patients** before exclusions.
- Excluded ($n = 268$): missing covariate/event-time information ($n = 244$), simultaneous relapse and death ($n = 21$), relapse at diagnosis ($n = 3$).
- **Final analytic cohort:** $N = 970$ **patients.**

Multi-state structure (illness–death, $T = 3$). Observed events: $1 \to 2$ relapse (307), $1 \to 3$ death w/o relapse (102), $2 \to 3$ death after relapse (188).

Covariates ($P = 7$): age, ER status, histological grade (II/III), nodal status, tumor size, treatment.

Models applied. All five Bayesian specifications: Shared-γ, Shared-$\mathcal{N}$, TS-γ, TS-$\mathcal{N}$, TS-MVN (Stan/HMC; 4 chains, 1,000 warmup + 1,000 sampling).

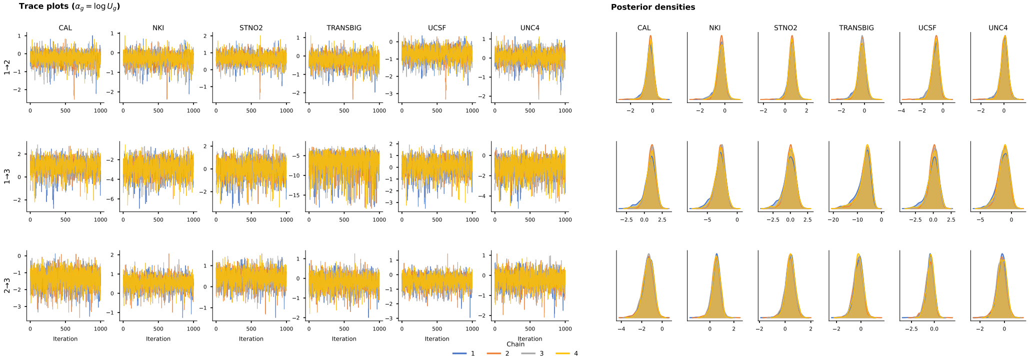
Real-world application: frailty variance estimates

Model	Transition	$\widehat{\mathrm{Var}}$	95% CrI	$\mathrm{ELPD}_{\mathrm{LOO}}$	ΔELPD (SE)
Shared-γ	All	0.31	(0.06, 1.01)	−3196.5	−86.4 (13.5)
Shared-$\mathcal{N}$	All	0.29	(0.05, 1.03)	−3196.7	−86.6 (13.5)
TS-γ	$1 \to 2$	0.43	(0.08, 1.31)	**−3110.1**	0.0 (ref)
	$1 \to 3$	**2.23**	(0.96, 4.47)		
	$2 \to 3$	0.61	(0.16, 1.66)		
TS-$\mathcal{N}$	$1 \to 2$	0.42	(0.07, 1.53)	−3111.0	−0.9 (0.5)
	$1 \to 3$	**3.94**	(1.40, 8.81)		
	$2 \to 3$	0.76	(0.15, 2.55)		
TS-MVN	$1 \to 2$	0.46	(0.08, 1.65)	−3111.6	−1.4 (0.6)
	$1 \to 3$	**4.15**	(1.43, 9.17)		
	$2 \to 3$	0.79	(0.16, 2.46)		
TS-MVN between-transition correlations					
	$\hat{\rho}_{21}$	0.07	$(-0.62, 0.74)$		
	$\hat{\rho}_{31}$	0.20	$(-0.51, 0.79)$		
	$\hat{\rho}_{32}$	−0.21	$(-0.82, 0.53)$		

- $\hat{\theta}_{1\to3}$ is 5–10× larger across all TS models ⇒ **transition-specific frailty strongly supported**.
- TS-MVN correlations: all 95% CrI $\ni 0$ ⇒ transitions approximately independent.
- **LOO-CV**: TS-γ best; Shared models far worse ($\gg 2\times$SE).

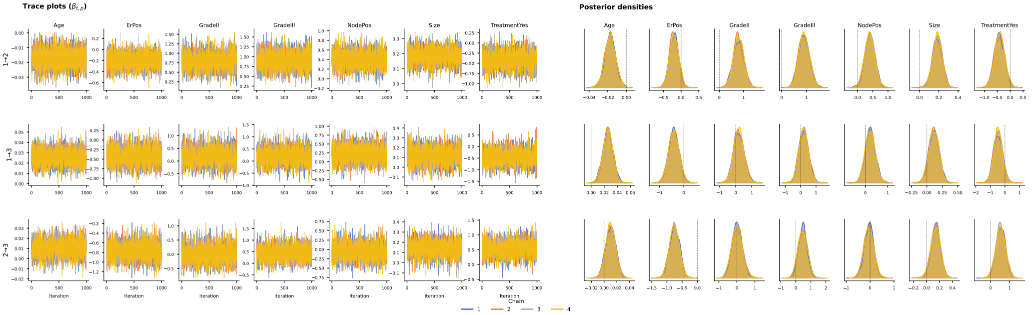
CrI: credible interval; $\mathrm{ELPD}_{\mathrm{LOO}}$: leave-one-out predictive score. 4 chains, 1,000 iter; $\hat{R} < 1.003$, ESS>2,800.

Posterior convergence: cluster frailties (TS-γ)



All 4 chains mix well with no drift. The $1 \to 3$ row shows wider, cluster-varying distributions: consistent with $\hat{\theta}_{1\to 3} = 2.23$, the largest frailty variance.

Posterior convergence: regression coefficients (TS-γ)



Chains overlap throughout; no divergences or treedepth warnings. $\hat{R} < 1.003$, ESS>2,800 for all parameters: supports reliability of the HR estimates on the next slide.

Real-world application: transition-specific hazard ratios

Posterior mean HR and 95% CrI (TS-γ model, LOO best):

Covariate	$1 \to 2$ (relapse)	$1 \to 3$ (death)	$2 \to 3$ (post-relapse)
Age	0.98 (0.97, 0.99)*	1.03 (1.01, 1.04)*	1.01 (1.00, 1.03)
ER positive	0.82 (0.63, 1.04)	0.68 (0.46, 0.99)*	0.48 (0.35, 0.62)*
Grade II	2.27 (1.49, 3.43)*	1.25 (0.66, 2.09)	1.06 (0.58, 1.88)
Grade III	2.53 (1.69, 3.69)*	1.26 (0.68, 2.18)	1.68 (0.86, 2.88)
Node positive	1.51 (1.07, 2.02)*	1.27 (0.83, 1.95)	0.97 (0.66, 1.36)
Tumor size	1.20 (1.10, 1.31)*	1.12 (0.96, 1.30)	1.15 (1.02, 1.31)*
Treatment	0.68 (0.47, 0.93)*	0.66 (0.37, 1.17)	1.76 (1.04, 2.58)*

- **1$\to$2**: Grade$\uparrow$, node-positive, larger tumor $\uparrow$ relapse; treatment protective (HR$= 0.68^*$).
- **1$\to$3**: Older age $\uparrow$ death risk; ER positivity protective (HR$= 0.68^*$).
- **2$\to$3**: ER positivity protective (HR$= 0.48^*$); treatment *increases* post-relapse mortality (HR$= 1.76^*$).

*95% CrI excludes 1. 4,000 draws; $\hat{R} < 1.003$.

Sensitivity to the prior on β: full results

The selected model (TS-γ) is refit with $\beta_{t,p} \sim N(0, s^2)$, $s \in \{1, 2.5, 5\}$ (frailty priors unchanged). Entries: posterior mean (95% CrI).

		$N(0,1)$	$N(0,2.5)$	$N(0,5)$
Panel A: Frailty variance				
Transition	$1 \rightarrow 2$	0.43 (0.08, 1.31)	0.43 (0.09, 1.33)	0.43 (0.09, 1.25)
	$1 \rightarrow 3$	**2.23** (0.96, 4.47)	**2.25** (0.93, 4.64)	**2.28** (0.96, 4.58)
	$2 \rightarrow 3$	0.61 (0.16, 1.66)	0.65 (0.16, 1.84)	0.65 (0.17, 1.84)
Panel B: Hazard ratios (covariates with 95% CrI excluding 1)				
$1 \rightarrow 2$	Age	0.98 (0.97, 0.99)	0.98 (0.97, 0.99)	0.98 (0.97, 0.99)
	Grade II	2.26 (1.46, 3.38)	2.41 (1.54, 3.71)	2.45 (1.54, 3.85)
	Grade III	2.52 (1.64, 3.80)	2.69 (1.71, 4.17)	2.72 (1.69, 4.23)
	Node pos.	1.51 (1.09, 2.05)	1.54 (1.11, 2.10)	1.55 (1.11, 2.10)
	Tumour size	1.20 (1.09, 1.32)	1.20 (1.09, 1.32)	1.20 (1.09, 1.32)
	Treatment	0.68 (0.45, 0.98)	0.66 (0.45, 0.95)	0.66 (0.44, 0.95)
$1 \rightarrow 3$	Age	1.03 (1.01, 1.04)	1.03 (1.01, 1.04)	1.03 (1.01, 1.04)
	ER pos.	0.68 (0.44, 1.01)	0.67 (0.43, 1.01)	0.67 (0.43, 1.01)
$2 \rightarrow 3$	ER pos.	0.48 (0.34, 0.65)	0.47 (0.34, 0.63)	0.47 (0.33, 0.63)
	Tumour size	1.15 (1.00, 1.31)	1.15 (0.99, 1.31)	1.15 (1.00, 1.31)
	Treatment	1.76 (1.05, 2.75)	1.84 (1.09, 2.93)	1.84 (1.06, 2.99)

Sensitivity to the prior on β: reading the results

- **Frailty variance is essentially invariant**: the $1 \to 3$ estimate stays far larger than the other two transitions under all three priors.
- **Regression effects are robust**: median change in posterior-mean log-HR is 0.02; the largest shift is $\Delta\mathrm{HR} = 0.20$ (Grade III, $1 \to 2$), a weakly-identified grade term more shrunk under the tighter $N(0, 1)$ (expected regularization, not instability). Well-identified effects (age, ER, size, treatment) are unchanged to two decimals.
- **No covariate changes significance or sign** across the three priors; the frailty-variance ordering ($1 \to 3 \gg 1 \to 2$, $2 \to 3$) is preserved throughout.
- **Frailty-variance prior** also varied: Gamma$(1, 1)$, Gamma$(0.1, 0.1)$, Gamma$(4, 2)$; the elevated $1 \to 3$ variance and all HR conclusions persist.

Summary

Contribution. We treat shared / TS-independent / TS-correlated frailty as *competing candidates*, fit by HMC and compared by LOO-CV, letting a simpler structure win when the data don't need more.

What the data show. Centers disagree almost entirely on **death without relapse**, pointing to non-cancer mortality, not tumor biology. Treatment also cuts both ways: it lowers relapse risk (HR$=0.68$) but raises post-relapse mortality (HR$=1.76$).

Caveat. With only six centers, near-zero correlations are *inconclusive*, not evidence of independence. In fact, even with twenty clusters in the simulation, real correlations still shrink toward 0, and we only had six centers here.

Bottom line

In clustered semi-competing risks, the key question is not *which frailty distribution to assume* but *whether heterogeneity is allowed to differ across transitions*.

References

Semi-competing risks & multi-state models

Fine, Jiang, Chappell (2001). *Biometrika* 88, 907–919.

Xu, Kalbfleisch, Tai (2010). *Biometrics* 66, 716–725.

Putter, Fiocco, Geskus (2007). *Stat. Med.* 26, 2389–2430.

Frailty models

Ripatti, Palmgren (2000). *Biometrics* 56, 1016–1022.

Lee, Dominici, Schrag, Haneuse (2016). *J. Am. Stat. Assoc.* 111, 1075–1095.

Ha, Xiang, Peng, Jeong, Lee (2020). *Lifetime Data Anal.* 26, 109–133.

Alvares, Haneuse, Lee, Lee (2019). `SemiCompRisks`. *R J.* 11, 376–400.

Bayesian computation & diagnostics

Hoffman, Gelman (2014). NUTS. *J. Mach. Learn. Res.* 15, 1593–1623.

Carpenter et al. (2017). Stan. *J. Stat. Softw.* 76, 1–32.

Vehtari, Gelman, Gabry (2017). LOO-CV & WAIC. *Stat. Comput.* 27, 1413–1432.

Lewandowski, Kurowicka, Joe (2009). LKJ prior. *J. Multivar. Anal.* 100, 1989–2001.

Vehtari, Gelman, Simpson, Carpenter, Bürkner (2021). $\hat{R}$. *Bayesian Anal.* 16, 667–718.

Breast cancer cohort

Haibe-Kains et al. (2012). *J. Natl. Cancer Inst.* 104, 311–325.

Thank You

Questions and comments are welcome.

Jinheum Kim

`jhkim@suwon.ac.kr`

33rd International Biometric Conference (IBC 2026)

July 12–16, 2026   Seoul, South Korea

Slides (Google Drive)